Sungil Kim

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RESEARCH INTEREST

Quality engineering and management, Time series data representation, Statistical process monitoring and anomaly detection, Industrial statistics and data analytics, AI+manufacturing/logistics

ACADEMIC APPOINTMENT

Ulsan National Institute of Science and Technology, Ulsan, Korea

Head, Department of Industrial Engineering	September 2024-present
Professor, Department of Industrial Engineering	September 2026-present
Associate Professor, Department of Industrial Engineering	September 2020-August 2026
Adjunct Professor, Artificial Intelligence Graduate School	September 2020-present
Director, Industry Intelligentization Institute	February 2025-present
Director, Center for Maritime Data Science	February 2019-April 2021
Assistant Professor, School of Management Engineering	July 2016-August 2020
Director, Center for Advanced Analytics	March 2018-February 2019

EDUCATION

Georgia Institute of Technology, Atlanta, Georgia, USA
H. Milton Stewart School of Industrial and Systems Engineering

Ph.D. in Industrial Engineering (with specialization in Statistics)	December 2011
Minor: Supply Chain Management	
M.S. in Statistics	December 2007
M.S. in Industrial Engineering	May 2007

Yonsei University, Seoul, Korea
Department of Computer Science and Industrial Systems Engineering

B.S. in Industrial Engineering, Magna Cum Laude	February 2005

Hansung Science High School, Seoul, Korea	February 1998

EMPLOYMENT

Outside Director, **Kakao Impact Foundation**	June 2026-May 2028
Full/Associate/Assistant Professor, **UNIST**	July 2016-present
Senior Engineer, **Samsung SDS**	January 2014-June 2016
Consultant, **Terra Technology**	September 2011-December 2013
Research/Teaching Assistant, **Georgia Institute of Technology**	May 2006-August 2011

PUBLICATIONS

JOURNAL PAPERS or REFEREED CONFERENCES (Under Review)

1. Seungsu Kam, Taewon Kang, and Sungil Kim**, ASTRA: Unsupervised Fault Localization via Attention Shift in Time-Series using iTransformer, under review.
2. Yongjin Cho, Minji Kim, and Sungil Kim**, Alert2Source: Multimodal Evidence-Grounded Synthesis for Air Quality Root-Cause Reporting, under review.
3. Junmok Kim, Kyubo Shin, Yungmin Sunwoo, Jongchan Kim, Sungil Kim, Jae Hoon Moon, and Gi-Soo Kim, Feature Transformation for Supervised Domain Adaptation Using Cross-Domain Paired Dataset, under review.
4. Chanbeom Hur and Sungil Kim**, The Grace Period Kaplan-Meier (GP-KM): A Consistent Survival Estimator for Customer Retention, under review.
5. YongKyung Oh, Sungil Kim**, and Alex A. T. Bui, Adaptive View Fusion for Human Activity Recognition with Heterogeneous Sensor Modalities, under review.
6. Seungsu Kam, Kyubo Shin, Wonsang Yoo, and Sungil Kim**, EU-Surv: Event Uncertainty-Aware Deep Dynamic Survival Modeling for Personalized Risk Prediction under Competing Risks, under review.
7. Seungsu Kam, Min-Su Shin*, Jitae Yoo, and Sungil Kim, Machine Learning Method to Detect Changing States in Quasar Light Curves. I. Online Analysis of Single-band Light Curves for Stochastic Changing States, under review.
8. Yongmin Kim, Byeonghoon Jeon, and Sungil Kim**, Rarity-Gated Context Conditioning for Offline Imitation Learning-Based Maritime Anomaly Detection, under review.
9. Seongjin Kim and Sungil Kim**, Multi-Field Hybrid Retrieval-Augmented Generation for Maritime Accident Root Cause Analysis, under review.

JOURNAL PAPERS

1. YongKyung Oh, JiIn Kwak, and Sungil Kim** (2026), Predicting non-recurrent congestion impact: A pattern-based approach for speed drop ratio prediction using weighted K-nearest Neighbors, *Computers & Industrial Engineering*, 123, pp 111769.
2. YongKyung Oh, Heeyoung Kim, and Sungil Kim** (2025), TSSI: Time Series as Screenshot Images for multivariate time series classification using convolutional neural networks, *Computers & Industrial Engineering*, 209, pp 111393.
3. YongKyung Oh, Giheon Koh, JiIn Kwak, Kyubo Shin, Gi-Soo Kim, Min Joung Lee, Hokyung Choung, Namju Kim, Jae Hoon Moon, and Sungil Kim** (2025), TAOD-Net: a data-driven approach for detecting Thyroid-Associated Orbitopathy symptoms using facial images, *Computers & Industrial Engineering*, 203, pp 111024.
4. YongKyung Oh, Kwonin Yoon, Jaemin Park, and Sungil Kim** (2024), Comparative evaluation of VAE-based monitoring statistics for real-time anomaly detection in AIS data, *Maritime Policy & Management*, 52(4), pp 609-626.
5. YongKyung Oh and Sungil Kim** (2024), Multi-modal lifelog data fusion for improved human activity recognition: a hybrid approach, *Information Fusion*, 110, pp 102464.

6. YongKyung Oh and Sungil Kim** (2023), Grid-Based Bayesian Bootstrap Approach for Real-Time Detection of Abnormal Vessel Behaviors from AIS Data in Maritime Logistics, *IEEE Transactions on Automation Science and Engineering*, 21(4), pp 6680-6692.
7. YongKyung Oh, Chiehyeon Lim, Junghye Lee, Sewon Kim, and Sungil Kim** (2024), Multi-channel convolution neural network for gas mixture classification, *Annals of Operations Research*, 339(1), pp 261-295.
8. Jonghwan Mun, Jitae Yoo, Heesun Kim, Nayi Ryu, and Sungil Kim** (2024), Domain knowledge-informed functional outlier detection in the refrigerator inspection lanes, *Computers & Industrial Engineering*, 189, pp 109936.
9. Junsub Jung, Sungil Kim*, and Heeyoung Kim (2023), Spatially-correlated time series clustering using location-dependent Dirichlet process mixture model, *Statistical Analysis and Data Mining*, accepted.
10. YongKyung Oh, Juhui Lee, and Sungil Kim** (2023), Sensor drift compensation for gas mixture classification in batch experiments, *Quality and Reliability Engineering International*, 39(6), pp 2422-2437.
11. Yeram Kim, Chiehyeon Lim*, Junghye Lee, Sungil Kim, Sewon Kim, and Dong-Hwa Seo (2023), Chemistry-informed machine learning: Using chemical property features to improve gas classification performance, *Chemometrics and Intelligent Laboratory Systems*, 237, 104808.
12. JuYeong Lee, JiIn Kwak, YongKyung Oh, and Sungil Kim** (2023), Quantifying incident impacts and identifying influential factors on urban traffic networks, *Transportmetrica B: Transport Dynamics*, 11(1), pp 279-300.
13. YongKyung Oh, JiIn Kwak, JuYeong Lee, and Sungil Kim** (2023), Time delay estimation of traffic congestion propagation due to accidents based on statistical causality, *Electronic Research Archive*, 31(2), pp 691-707.
14. Sungil Kim* (2021), Maximum feasibility estimation, *Information Sciences*, 575, pp 739-801.
15. Sungil Kim, Rong Duan, Guang-Qin Ma, and Heeyoung Kim* (2020), Multiresolution spatial generalized linear mixed model for integrating multi-fidelity spatial count data without common identifiers between data sources, *Spatial Statistics*, 39, 100467.
16. Sungil Kim, Heeyoung Kim*, and Jye-Chyi Lu (2019), A practical approach to measuring the impacts of stockouts on demand, *Journal of Business and Industrial Marketing*, 34(4), pp 891-901.
17. Sungil Kim* (2019), Revealing household characteristics using connected home products, *Information Sciences*, 486, pp 52-61.
18. Heeyoung Kim, Rong Duan*, Sungil Kim, Jaehwan Lee, and Guang-Qin Ma (2019), Spatial cluster detection in mobility networks: a copula approach, *Journal of the Royal Statistical Society: Series C*, 68(1), pp 99-120.
19. Heeyoung Kim, Justin T. Vastola, Sungil Kim, Jye-Chyi Lu*, and Martha A. Grover (2017), Incorporation of engineering knowledge into the modeling process: a local approach, *International Journal of Production Research*, 55(20), pp 5865-5880.

20. Heeyoung Kim, Sungil Kim, Jian Deng, Jye-Chyi Lu*, K. Wang, C. Zhang, Martha A. Grover, and B. Wang (2017), An integrated holistic model of a complex process, *International Journal of Advanced Manufacturing Technology*, 89(1), pp 1137-1147.

21. Sungil Kim, Heeyoung Kim*, and Yongro Park (2017), Early detection of vessel delays using combined historical and real-time information, *Journal of the Operational Research Society*, 68(2), pp 182-191.

22. Heeyoung Kim, Justin T. Vastola, Sungil Kim*, Jye-Chyi Lu, and Martha A. Grover (2017), Batch sequential minimum energy design with design region adaptation, *Journal of Quality Technology*, 49(1), pp 11-26.

23. Sungil Kim, Heeyoung Kim*, and Younghwan Namkoong (2016), Ordinal classification of imbalanced data with application in emergency and disaster information services, *IEEE Intelligent Systems*, 31(5), pp 50-56.

24. Sungil Kim and Heeyoung Kim* (2016), A new metric of absolute percentage error for intermittent demand forecasts, *International Journal of Forecasting*, 32(3), pp 669-679.

25. Sungil Kim, Heeyoung Kim*, Richard W. Lu, Jye-Chyi Lu, Michael J. Casciato, and Martha A. Grover (2015), Adaptive combined space-filling and D-optimal designs, *International Journal of Production Research*, 53(17), pp 5354-5368.

26. Sungil Kim, Heeyoung Kim*, Jye-Chyi Lu, Michael J. Casciato, Martha A. Grover, Dennis W. Hess, Richard W. Lu, and Xin Wang (2015), Layers of experiments with adaptive combined design, *Naval Research Logistics*, 62(2), pp 127-142.

REFEREED CONFERENCE PAPERS

1. Jonghun Lee, YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2026), Continuum Dropout for Neural Differential Equations. *Proceedings of the AAAI conference on artificial intelligence. Vol.40.*

2. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2026), FlowPath: Learning Data-Driven Manifolds with Invertible Flows for Robust Irregularly-sampled Time Series Classification. *Proceedings of the AAAI conference on artificial intelligence. Vol.40.*

3. YongKyung Oh, Dong-Young Lim, Sungil Kim**, and Alex A. T. Bui (2025), TANDEM: Temporal Attention-guided Neural Differential Equations for Missingness in Time Series Classification. *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM).*

4. YongKyung Oh, Seungsu Kam, Jonghun Lee, Dong-Young Lim, and Sungil Kim** (2025), Comprehensive Review of Neural Differential Equations for Time Series Analysis. **Survey Paper**, *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI-25).*

5. YongKyung Oh, Seungsu Kam, Dong-Young Lim, and Sungil Kim** (2025), Modeling Irregular Astronomical Time Series with Neural Stochastic Delay Differential Equations. **Short Paper**, *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM).*

6. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2025), Neural Differential Equations for Continuous-Time Analysis. **Tutorial**, *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM).*

7. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2025), DualDynamics: Synergizing Implicit and Explicit Methods for Robust Irregular Time Series Analysis. *Proceedings of the AAAI conference on artificial intelligence. Vol.39.*

8. YongKyung Oh, Dong-Young Lim, Sungil Kim, and Alex Bui (2024), Attention-guided Neural Differential Equations Framework for Missingness in Time series Classification. *The 10th Mining and Learning from Time Series Workshop: From Classical Methods to LLMs (KDD MILETS Workshop 2024).*

9. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2024), Neural Langevin-type Stochastic Differential Equations for Astronomical time series Classification under Irregular Observations. *In ICLR 2024 Workshop on AI4DifferentialEquations In Science.*

10. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2024), Stable neural stochastic differential equations for irregular time series classification, *The Twelfth International Conference on Learning Representations (ICLR) (**spotlight**).*

11. YongKyung Oh and Sungil Kim** (2021), Multi-channel convolution neural network for gas mixture classification, *2021 International Conference on Data Mining Workshops (ICDMW)*, pp. 1094-1095.

12. YongKyung Oh and Sungil Kim** (2021), Multichannel convolution neural network for gas mixture classification, *The 4th Conference of Korean Artificial Intelligence Association, Online.*

13. YongKyung Oh and Sungil Kim** (2019), Exploiting Logistics Anomaly Detection using Maritime Big Data. In *IIE Annual Conference. Proceedings*, pp. 964-969, Institute of Industrial and Systems Engineers (IISE).

14. Juhui Lee and Sungil Kim** (2019), Ordinal-imbalanced Data Classification through Data Reduction by Singular Value Decomposing Truncation. In *IIE Annual Conference. Proceedings*, pp. 236-256, Institute of Industrial and Systems Engineers (IISE) [***2019 IISE Best Paper Award, Quality Control & Reliability Engineering Division***].

15. Michael J. Casciato, Sungil Kim, Jye-Chyi Lu, Dennis W. Hess, Martha A. Grover (2012), Optimization of carbon dioxide-assisted nanoparticle deposition process with uncertain design space, *Proceedings of the 11th International Symposium on Process Systems Engineering*, pp 1191-1195.

OTHER PUBLICATIONS: DOMESTIC JOURNALS & BOOK CHAPTERS

1. Seongjin Kim and Sungil Kim** (2026), A Hybrid Prompt Agent for Maritime AIS Data Analysis: Performance Optimization through Query Classification and Dynamic Routing, *Journal of the Korean Institute of Industrial Engineers*, accepted

2. YongKyung Oh, Sungil Kim, and Alex AT Bui (2024, August), Deep Interaction Feature Fusion for Robust Human Activity Recognition. *In International Joint Conference on Artificial Intelligence* (pp. 99-116). Singapore: Springer Nature Singapore.

3. YongKyung Oh, Hanjoo Kim, Deokjung Lee, and Sungil Kim** (2021), Simulation-based anomaly detection in nuclear reactors, *Journal of the Korean Institute of Industrial Engineers*, 47(2), pp 130-143.
4. YongKyung Oh, Namu Kim, and Sungil Kim** (2021), Transfer Learning based approach for mixture gas classification, *Journal of the Korean Institute of Industrial Engineers*, 47(2), pp 144-159.
5. Jaemin Park and Sungil Kim** (2020), Maritime anomaly detection based on VAE-CUSUM monitoring system, *Journal of the Korean Institute of Industrial Engineers*, 46(4), pp 432-442.
6. YongKyung Oh and Sungil Kim** (2018), Research trends for solving the odor problem, *Industrial Engineering Magazine*, 25(3), 45-51.
7. Seung-Kweon Hong*, Sungil Kim (2005), A time prediction model of cursor movement with path constraints, *Journal of the Korean Institute of Industrial Engineers*, 31(4), pp 334-340.

INDIVIDUAL STUDENT GUIDANCE

1. Hyejin Cho, MS. Started in Spring 2023; passed defense on December 17, 2025; thesis title: "Assessing User Retention of Updatable Home Appliances: Survival Analysis;" graduated in February 2026; first position: Hanjin Logistics Institute
2. Seongjin Kim, MS. Started in Spring 2024; passed defense on December 15, 2025; thesis title: "A Retrieval-Augmented Generation Framework for Root Cause Analysis in Maritime Accident Adjudication Reports;" graduated in February 2026; first position: BNK Kyongnam Bank
3. Taewon Kang, MS. Started in Fall 2023; passed defense on June 9, 2025; thesis title: "An Unsupervised Framework for Online Failure Prediction and Fault Diagnosis in Industrial Systems;" graduated in August 2025; first position: Posco Future M
4. Jae Hun Cho, MS. Started in Spring 2023; passed defense on December 17, 2024; thesis title: "An Iterative Framework for Active Domain Adaptation in Heterogeneous Image Streams;" graduated in August 2025
5. Byungkook Koo, MS. Started in Spring 2022; passed defense on June 11, 2024; thesis title: "Enhancing Training Data Quality in Real-Time Streams: An Iterative Framework for Adaptive Domain Adaptation;" graduated in August 2024
6. Heesun Kim, MS. Started in Spring 2022; passed defense on December 4, 2023; thesis title: "Improving Quality Inspection Performance in Refrigerator Manufacturing Systems through Sensor Calibration and Functional Outlier Detection;" graduated in February 2024; first position: LG Electronics
7. Jitae Yoo, MS. Started in Fall 2021; passed defense on June 4, 2023; thesis title: "Enhanced Deep Anomaly Detection In Contaminated Datasets Using Semi-Supervised Learning;" graduated in August 2023; first position: LX Hausys
8. JiIn Kwak, MS. Started in Spring 2021; passed defense on June 4, 2023; thesis title: "Graph Neural Networks for Traffic Incident Congestion Detection and Impact Prediction;" graduated in August 2023; first position: LG Electronics

9. YongKyung Oh, PhD. Started in Spring 2018; passed defense on December 9, 2022; thesis title: "A Comprehensive Study of Deep Learning for Real-World Multivariate Time Series Classification;" graduated in February 2023; first position: Postdoc at UCLA
10. Jonghwan Mun, MS. Started in Spring 2021; passed defense on December 7, 2022; thesis title: "Domain Knowledge-Informed Functional Outlier Detection for Line Quality Control Systems;" graduated in February 2023; first position: LG Electronics
11. Gyeongjun Kim, MS. Started in Spring 2021; passed defense on December 7, 2022; thesis title: "Semi-Supervised Contrastive Learning for Anomaly Detection in Contaminated Data;" graduated in February 2023; first position: CJ Logistics
12. Giheon Koh, MS. Started in Spring 2021; passed defense on November 29, 2022; thesis title: "Real time Risk Assessment based on Irregularly Sampled Time Series Data from Wearable Devices;" graduated in February 2023; first position: LG Display
13. Kwonin Yoon, MS. Started in Fall 2020; passed defense on June 15, 2022; thesis title: "Vessel Behavior Assessment and Anomaly Detection from AIS Data;" graduated in August 2022; first position: LG Electronics
14. JuYeong Lee, MS. Started in Fall 2019; passed defense on June 17, 2021; thesis title: "Modelling Traffic Incidents Impacts and Prediction of Incident driven Congestion Propagation in a Large Scale Road Network;" graduated in August 2021; first position: Researcher at QRAFT
15. Jaemin Park, MS. Started in Spring 2018; passed defense on December 16, 2020; thesis title: "Unsupervised Anomaly Detection to Monitor Delays of Vessel Voyages based on Variational Autoencoder and Control Chart;" graduated in February 2021; first position: CEO at THYROSCOPE Inc.
16. Juhui Lee, MS. Started in Fall 2018; passed defense on June 18, 2020; thesis title: "Data Quality Improvements for Multiclass Classification;" graduated in August 2020; first position: Senior Researcher at THYROSCOPE Inc.

PATENTS

1. Kim, Sungil (primary inventor), Apparatus and method for human activity recognition through hybrid fusion of dynamic and static data. (10-2932931, granted February 25, 2026)(KR 10-2024-0033382, applied March 8, 2024).
2. Kim, Sungil (co-inventor), Apparatus for generating target gas detection model based on CNN learning and method thereof. (10-2811004, granted May 16, 2025)(KR 10-2021-0135617, applied October 13, 2021).
3. Kim, Sungil (primary inventor), Method, system, and operating method for measuring customer retention rate based on survival analysis using log data. (KR 10-2025-0024769, applied February 26, 2025).
4. Kim, Sungil (primary inventor), Method, computer device, and computer program to predict road congestion propagation using pattern matching. (10-2608343, granted November 27, 2023)(KR 10-2021-0120229, applied September 9, 2021)(JP 2022-137508, applied August 31, 2022).

5. Kim, Sungil (primary inventor), Method, computer device, and computer program to predict propagation time delay lag of road congestion using transfer entropy. (10-2604575, granted November 26, 2023)(KR 10-2021-0119772, applied September 8, 2021)(JP 2022-137509, applied August 31, 2022).
6. Kim, Sungil (primary inventor), Method of anomaly detection of vessels applying Bayesian bootstrap. (10-2534357, granted May 16, 2023)(KR 10-2020-0178651, applied December 18, 2020).
7. Kim, Sungil (primary inventor), Encoding apparatus and method for converting time series data into images. (KR 10-2022-0183341, applied December 22, 2022).
8. Kim, Sungil (primary inventor), Sensor drift compensation method and device. (US 18/012,626, applied December 22, 2022)(10-2364019, granted February 14, 2022)(PCT/KR2021/010729, applied August 12, 2021)(KR 10-2020-0101037, applied August 14, 2020).
9. Kim, Sungil (primary inventor), Method and apparatus for determining delay possibility of shipment. (10-2250354, granted May 4, 2021), (PCT/KR2020/015854, applied November 12, 2020).
10. Kim, Sungil (primary inventor), Apparatus and method for sensor based realtime odor classification. (10-2106561, granted April 24, 2020), (China 201910234530.5, applied March 26, 2019).
11. Kim, Sungil (primary inventor), Method for risk scoring in logistics system. (KR 10-2015-0150296, applied October 28, 2015).
12. Kim, Sungil (primary inventor), System and method for grid-based geofencing service, (KR 10-2015-0090396, applied June 25, 2015).
13. Kim, Sungil (co-inventor), System and method for detecting and predicting anomalies based on analysis of text data, (KR 10-2014-0142784, applied October 21, 2014).
14. Kim, Sungil (co-inventor), System and method for detecting and predicting anomalies based on analysis of time-series data. (USA 09952921, granted April 24, 2018), (KR 10-2014-0136765, applied October 10, 2014).
15. Kim, Sungil (primary inventor), Apparatus and method for calculating standard route of moving body. (10-1766640, granted August 3, 2017), (KR 10-2014-0127859, applied September 24, 2014).
16. Kim, Sungil (primary inventor), Apparatus and method for early detection of abnormality. (KR 10-2014-0112321, applied August 27, 2014).

GRANTS & CONTRACTS

1. Multimodal Foundation Model-Driven Manufacturing, Predictive Maintenance, and Process Optimization
   - Role: Participant
   - Source: IITP
   - Duration: April 1, 2026-December 31, 2030
   - Amount: ₩4,515,000,000

2. A Study on Quantitative Quality Assessment of Transit Wayfinding Routes in Map Services
   - Role: Principal Investigator
   - Source: NAVER
   - Duration: April 8, 2026-December 31, 2027
   - Amount: ₩150,000,000
3. The AI Era, A Knowledge Platform for All: Libraries as a Social Operating System for Sustainable Growth
   - Role: Group Leader
   - Source: Global Humanities and Social Sciences Convergence Research Support Project, National Research Foundation of Korea (NRF)
   - Duration: September 1, 2025-August 31, 2030
   - Amount: ₩10,000,000,000
4. Ulsan Industry-University Convergence Zone: Enterprise-Focused Industry-University Convergence Promotion Support
   - Role: Principal Investigator
   - Source: Industry-University Convergence District Creation Project, Korea Institute for Advancement of Technology (KIAT)
   - Duration: January 1, 2025-December 31, 2027
   - Amount: ₩735,000,000
5. Development of Domain-free Time Series Data Drift Management Techniques
   - Role: Principal Investigator
   - Source: Mid-Career Researcher Program, National Research Foundation of Korea (NRF)
   - Duration: March 1, 2025-February 29, 2028
   - Amount: ₩650,000,000
6. Development of Foundation Models for Bioelectrical Signal Data and Validation of Their Clinical Applications: A Noise-and-Variability Robust, Generalizable Self-Supervised Learning Approach
   - Role: Participant
   - Source: IITP
   - Duration: July 1, 2024-December 31, 2026
   - Amount: ₩2,760,000,000
7. Human Centered-Carbon Neutral Global Supply Chain Research Center
   - Role: Group Leader (Data Science Group)
   - Source: Engineering Research Center (ERC), National Research Foundation of Korea (NRF)
   - Duration: June 1, 2023-May 31, 2030

- Amount: ₩13,500,000,000

8. Enhancement of Fault Detection/Discrimination Models in LQC Processes
   - Role: Principal Investigator
   - Source: LG Electronics
   - Duration: June 30, 2023-June 28, 2024
   - Amount: ₩50,000,000
9. Developing data-driven user satisfaction metrics for ThinQ contents in LG Upgradable home appliances
   - Role: Principal Investigator
   - Source: LG Electronics
   - Duration: June 1, 2023-May 31, 2024
   - Amount: ₩80,000,000
10. Development of a robust methodology for thyroid eye disease recognition based on Heterogeneous devices
    - Role: Participant
    - Source: UNIST-THYROSCOPE Inc.
    - Duration: March 1, 2023-December 31, 2024
    - Amount: ₩200,000,000
11. Thyroid eye disease recognition based on Heterogeneous cameras
    - Role: Participant
    - Source: Korea Innovation Foundation
    - Duration: October 1, 2022-January 31, 2023
    - Amount: ₩60,000,000
12. Research on AI solutions for predicting appropriate thyroid drug dosage
    - Role: Principal Investigator
    - Source: TIPA(the Korea Technology and Information Promotion Agency for SMEs)
    - Duration: May 1, 2022-April 30, 2024
    - Amount: ₩75,864,000
13. Damage detection of linear compressors for refrigerators in line quality control system
    - Role: Principal Investigator
    - Source: LG Electronics
    - Duration: November 29, 2021-November 28, 2022
    - Amount: ₩50,000,000
14. AI Innovation Hub R&D
    - Role: Participant

- Source: IITP
- Duration: July 1, 2021-December 31, 2025
- Amount: ₩1,837,500,000

15. Thyroid eye disease recognition using convolutional neural networks
    - Role: Participant
    - Source: Korea Innovation Foundation
    - Duration: August 1, 2021-December 31, 2021
    - Amount: ₩61,000,000
16. Development of biosignal error correction model between wearable devices
    - Role: Principal Investigator
    - Source: Ulsan Industry University Convergence Institute
    - Duration: September 1, 2021-December 31, 2021
    - Amount: ₩19,400,000
17. Development of personalized self-management solutions and disease risk prediction techniques for thyroid disease management
    - Role: Participant
    - Source: UNIST AI Innovation Park
    - Duration: August 1, 2021-July 31, 2022
    - Amount: ₩100,000,000
18. Development of AI-based vessel ETA (Estimated Time of Arrival) prediction methodology for intelligent smart ports
    - Role: Principal Investigator
    - Source: National Research Foundation of Korea (NRF)
    - Duration: June 1, 2021-February 29, 2024
    - Amount: ₩131,343,000
19. Development of AI technology for identifying traffic congestion patterns
    - Role: Participant
    - Source: NAVER
    - Duration: November 1, 2020-October 31, 2022
    - Amount: ₩300,000,000
20. AI dataset construction for manufacturing facilities
    - Role: Participant
    - Source: the Korea Technology and Information Promotion Agency for SMEs (TIPA)
    - Duration: Sep 18, 2020-November 30, 2020
    - Amount: ₩283,739,091

21. Manufacturing data analysis and AI model development
    - Role: Participant
    - Source: interX
    - Duration: June 10, 2020-November 30, 2020
    - Amount: ₩122,727,273
22. Incorporation of domain knowledge into the modeling process for quality improvement in smart manufacturing
    - Role: Principal Investigator
    - Source: National Research Foundation of Korea (NRF)
    - Duration: March 1, 2017-February 28, 2022
    - Amount: ₩250,000,000
23. Structural analysis on the processes of technological innovation in the fourth industrial revolution: focusing on open innovation and concurrent transformation of regional innovation system
    - Role: Participant
    - Source: National Research Foundation of Korea (NRF)
    - Duration: July 1, 2018-June 30, 2021
    - Amount: ₩228,762,000
24. Development of AI-Based Reactor Core Diagnostics System
    - Role: Participant
    - Source: Korea Hydro & Nuclear Power Co.,Ltd.
    - Duration: June 1, 2019-May 31, 2021
    - Amount: ₩446,557,680
25. Development of stowage optimization engine solver using reinforcement learning
    - Role: Principal Investigator
    - Source: Cyberlogitec
    - Duration: November 1, 2018-August 31, 2020
    - Amount: ₩45,000,000
26. Development of Algorithms for Mixture Gases Classification
    - Role: Principal Investigator
    - Source: Ulsan Industry-University Convergence Campus
    - Duration: March 16, 2020-July 31, 2020
    - Amount: ₩40,000,000
27. Smart Port Logistics Support Center
    - Role: Principal Investigator

- Source: Ulsan Port Authority
- Duration: January 1, 2019-December 31, 2019
- Amount: ₩1,395,460,000

28. A Constraint satisfaction problem with attribute data: an application to connected home products
    - Role: Principal Investigator
    - Source: UNIST
    - Duration: September 1, 2016-August 31, 2019
    - Amount: ₩20,000,000
29. Data mining project lab
    - Role: Participant
    - Source: UNIST-Taesung Environmental Research Institute
    - Duration: October 1, 2018-December 31, 2018
    - Amount: ₩15,000,000
30. Development of data analytics methods for mixture gases classification
    - Role: Principal Investigator
    - Source: UNIST-Taesung Environmental Research Institute
    - Duration: August 1, 2018-December 31, 2018
    - Amount: ₩10,000,000
31. Blockchain-based system engineering
    - Role: Participant
    - Source: UNIST
    - Duration: March 1, 2018-December 31, 2018
    - Amount: ₩45,000,000
32. Development of data analytics methods to identify the sources of odor
    - Role: Principal Investigator
    - Source: Ulsan Industry-University Convergence Campus
    - Duration: October 1, 2017-March 31, 2018
    - Amount: ₩38,500,000
33. Development and application of methods and an intelligent platform system for Industry 4.0
    - Role: Participant
    - Source: UNIST
    - Duration: February 1, 2017-December 31, 2017
    - Amount: ₩40,000,000

TEACHING EXPERIENCE

Ulsan National Institute of Science and Technology, Ulsan, Korea
Department of Industrial Engineering
Instructor

- NOVA 505: Probability and Statistics for Safety and Risk Data — Spring, 2026
- IE 362/MGE 362: Statistical Quality Management — Spring, 2017-2026
- IE 552/AI 603: Neural Differential Equations — Fall, 2024
- IE 313: Time Series Analysis — Fall, 2023
- IE 509/AI 533: Advanced Quality Control — Fall, 2021-2022
- AI 590: AI Graduate Seminar — Spring, 2022
- AI 501: Introduction to AI — Spring, 2022
- IE 471: Special Topic (Project Lab) — Fall, 2021
- MGT 101: Entrepreneurship & Big Data — Fall, 2020
- IE 502/MGE 502: Statistical Programming — Fall, 2018-2020
- TIM 713: Industrial Innovation Seminar — Fall, 2018
- MGE 509: Advanced Quality Control — Fall, 2017-2019
- MGE 301: Operations Research I — Fall, 2016-2017
- MGT 209: Operations Management — Fall, 2016

Georgia Institute of Technology, Atlanta, Georgia, USA
H. Milton Stewart School of Industrial and Systems Engineering
Teaching Assistant

- ISyE 4031: Regression and Forecasting — Spring, 2011
- ISyE 3770: Probability and Statistics — Fall, 2010
- ISyE 2027: Probability with Applications — Spring, 2010
- ISyE 4803: Advanced Supply Chain Logistics — Fall, 2007
- ISyE 6739: Statistical Methods — Summer, 2007

PRESENTATIONS

INTERNATIONAL

1. Rarity-Gated Context Conditioning for Offline Imitation Learning-Based Maritime Anomaly Detection, *IEEE CASE 2026*, August 2026.
2. Multi-Field Hybrid Retrieval-Augmented Generation for Maritime Accident Root Cause Analysis, *IEEE CASE 2026*, August 2026.
3. Deep Learning Framework for Personalized Antithyroid Dose Recommendation Using Dual Survival Models, *IISE Annual Conference & Expo 2026*, May 2026.
4. Nonparametric Estimation of Customer Retention in Non-contractual Settings, *IISE Annual Conference & Expo 2026*, May 2026.

5. Nonparametric and Recurrent Survival Models for User Retention , *INFORMS Annual Meeting*, October 2025.
6. Assessing User Retention of Updatable Home Appliances: Survival Analysis, *IISE Annual Conference & Expo 2025*, May 2025.
7. Grid-Based Bayesian Bootstrap Approach for Real-Time Detection of Abnormal Vessel Behaviors from AIS Data in Maritime Logistics, *2024 IEEE 20th International Conference on Automation Science and Engineering*, August 2024.
8. (**Invited Talk**) Domain-Knowledge-Informed Functional Outlier Detection for Line Quality Control Systems, *2nd INFORMS Conference on Quality, Statistics and Reliability*, July 2024.
9. (**Invited Talk**) Integrating Neural Controlled Differential Equations and Neural Flow for Comprehensive Irregularly-sampled Time Series Analysis, *IISE Annual Conference & Expo 2024*, May 2024 [**2024 Best Track Paper Competition Finalist, Data Analytics and Information Systems (DAIS) Division**].
10. Optimal and Personalized Dose Determination for Patients with Thyroid Hormone Disorders Using Deep Learning-Based Survival Analysis, *IISE Annual Conference & Expo 2024*, May 2024.
11. Enhancing Human Activity Recognition with Comparative Data Fusion and Learning Strategies, *IISE Annual Conference & Expo 2024*, May 2024.
12. Enhancing Astronomical time series Classification with Neural Stochastic Differential Equations under Irregular Observations, Knowledge Guided ML (KGML) Bridge Program, *The 38th Annual AAAI Conference on Artificial Intelligence 2024* (Poster), February 2024.
13. Invertible Solution of Neural Differential Equations for Analysis of Irregularly-Sampled Time Series, Artificial Intelligence for Time Series (AI4TS) Workshop, *The 38th Annual AAAI Conference on Artificial Intelligence 2024* (poster), February 2024.
14. Unifying Neural Controlled Differential Equations and Neural Flow for Irregular Time Series Classification, *The Symbiosis of Deep Learning and Differential Equations Workshop, NeurIPS 2023 Workshop* (poster), November 2023.
15. (**Invited Talk**) Comparative evaluation of VAE-based monitoring statistics for real-time anomaly detection in AIS data, The 11th International Conference on Logistics and Maritime Systems (LOGMS 2023), September 2023.
16. Real-time risk assessment of thyroid function abnormality using irregularly-sampled heart rate records, *IISE Annual Conference & Expo 2023*, May 2023.
17. Sparse multi-channel convolutional neural network for multivariate time series classification, *IISE Annual Conference & Expo 2023*, May 2023.
18. TAOD-Net: Near-ophthalmologist level active thyroid-associated orbitopathy detection on frontal eye photographs with deep learning, *IISE Annual Conference & Expo 2023*, May 2023.
19. Domain knowledge-informed functional outlier detection for line quality control systems, *IISE Annual Conference & Expo 2023*, May 2023.
20. (**Invited Talk**) Neural stochastic differential equations based on the Ornstein-Uhlenbeck process, *IISE Annual Conference & Expo 2023*, May 2023 [**2023 Best Track Paper Competition Finalist, Data Analytics and Information Systems (DAIS) Division**].

21. (**Invited Talk**) Domain knowledge-informed functional outlier detection for line quality control systems, *Industrial and Systems Engineering Virtual Seminar*, Rutgers University, March 2023.
22. (**Invited Talk**) Time delay estimation of traffic congestion based on statistical causality, *3rd Workshop on Data-driven Intelligent Transportation* (DIT 2022), Held in conjunction with CIKM 2022, October 2022.
23. Sensor drift compensation for gas mixture classification in batch experiments, *IISE Annual Conference & Expo, May 2022*.
24. A propagation prediction method for non-recurrent traffic congestion, *IISE Annual Conference & Expo*, May 2022.
25. (**Invited Talk**) Time delay estimation of traffic congestion based on statistical causality, *The 5th International Conference on Econometrics and Statistics*(EcoSta 2022), June 2022.
26. (**Invited Talk**)Detection of abnormal vessel behaviors from AIS data using the Bayesian bootstrap, *Industrial and Systems Engineering Virtual Seminar*, Rutgers University, November 2021.
27. A propagation prediction method for non-recurrent traffic congestion, *INFORMS Annual Meeting*, October 2021.
28. VAE-CUSUM: a new feature-based monitoring chart to monitor vessel voyages using AIS data, *INFORMS Annual Meeting*, October 2021.
29. Multi-channel convolution neural network for gas mixture classification, *IISE Annual Conference & Expo 2021*, May 2021.
30. Logistics anomaly detection with maritime big data: a bootstrap approach, *IISE Annual Conference & Expo 2021*, May 2021.
31. Sensor drift compensation for mixed gas classification under batch experiments, *INFORMS Annual Meeting*, November 2020.
32. (**Invited Talk**) Maximum feasibility estimation, *The 2020 INFORMS Workshop on Data Science*, November 2020.
33. Logistics anomaly detection with maritime big data: a bootstrap approach, *IISE Annual Conference & Expo 2020*, November 2020.
34. Maximum feasibility estimation, *INFORMS Annual Meeting*, October 2019.
35. (**Invited Talk**) Revealing household characteristics using connected home products, *The Fifth International Conference on the Interface between Statistics and Engineering*, June 2019.
36. Ordinal-imbalanced data classification by singular value decomposing truncation, *IISE Annual Conference & Expo 2019*, May 2019.
37. Exploiting logistics anomaly detection using maritime big data, *IISE Annual Conference & Expo 2019*, May 2019.
38. Early detection of vessel delays using combined historical and real-time information, *INFORMS Annual Meeting*, October 2017.
39. Layers of experiments with adaptive combined design, *INFORMS Annual Meeting*, November 2014.

40. (**Invited Talk**) Prediction & inference using the hierarchical spatiotemporal varying coefficient model: applied to detailed sales forecasting for retail providers, *Georgia Tech Research and Innovation Conference*, February 2010.
41. Detailed sales forecasting & promotion analysis for retail providers, *Joint Statistical Meetings*, August 2009.

DOMESTIC

1. Semi-Supervised Defect Classification with Active Label Expansion for Perforated Packaging Inspection under Limited Annotations, KIIE Annual Spring Conference, 2026.
2. A Satellite-based Multimodal Deep Learning and Automated Diagnosis System for Air Pollution Source Identification, KIIE Annual Spring Conference, 2026.
3. (**Invited Talk**) Neural Differential Equations for Continuous-Time Modeling and Analysis, *Brain Korea 21 (BK21) Program Seminar*, Korea University, November 2025.
4. Learning from Patient Similarity: A Cluster-Based Approach to ECG Imputation, KIIE Annual Fall Conference, 2025.
5. Integrating Environmental Data into Decision-Making Based Maritime Anomaly Detection, KIIE Annual Fall Conference, 2025.
6. Missing Mechanism-Aware Diagnosis for Multivariate Time Series Imputation, November 2025.
7. Domain-Specific Foundation Models for Initial Bio-device Deployment, KIIE Annual Fall Conference, 2025.
8. Deep Dynamic Survival Modeling for Personalized Risk Prediction under Competing Risks, KIIE Annual Spring Conference, 2025.
9. Improving Ship Fuel Consumption Prediction via Environmental Data Integration, KIIE Annual Spring Conference, 2025.
10. Statistical Modeling of Rolling Retention via Recurrent Survival Analysis, KIIE Annual Spring Conference, 2025.
11. (**Invited Talk**) Stable Neural Stochastic Differential Equations for Irregular Time Series Classification, KDMS Conference, November 2024.
12. Addressing Spatial Gaps in XCO_2 Monitoring: A Machine Learning Approach for South Korea, KIIE Annual Fall Conference, 2024.
13. Personalized Initial Dosage Recommender System for Hyperthyroidism Patients Using Deep Survival Analysis, KIIE Annual Spring Conference, 2024.
14. Assessing User Retention of Upgradable Home Appliances: Survival Analysis, KIIE Annual Spring Conference, 2024.
15. Optimal and personalized dose determination for patients with thyroid hormone disorders using deep learning-based survival analysis, KIIE Annual Fall Conference, 2023.
16. An intelligent diagnostic system for thyroid-associated ophthalmopathy using images from heterogeneous smartphone devices, KIIE Annual Fall Conference, 2023.

17. (**Invited Talk**) Heterogeneous model fusion for enhanced sensor-based human activity recognition, Korea Computer Congress (KCC), June 2023 [***2023 KCC Best Presentation Award, AI Division***].
18. (**Invited Talk**) Flow-based neural differential equations for time series analysis, Korea Computer Congress (KCC), June 2023 [***2023 KCC Best Paper Award, AI Division***].
19. Spatio-temporal graph neural network approach for real-time traffic incident detection, KIIE Annual Spring Conference, 2023.
20. Flow-based neural differential equations for time series analysis, KIIE Annual Spring Conference, 2023.
21. Heterogeneous model fusion for enhanced sensor-based human activity recognition, KIIE Annual Spring Conference, 2023.
22. Real-time detection of changing-state active galactic nuclei using a deep learning approach, KIIE Annual Spring Conference, 2023.
23. (**Invited Talk**) Lifelog data fusion approach for emotion recognition, Conference of Korea Software Congress 2022 (KSC 2022), December 2022.
24. (**Invited Talk**) Irregularly sampled time series classification using neural stochastic differential equation, Conference of Korean Artificial Intelligence Association, November 2022.
25. (**Invited Talk**) Real-time risk assessment of thyroid function abnormality using irregularly-sampled heart rate records, Conference of Korean Artificial Intelligence Association, November 2022.
26. Irregularly sampled time series classification using neural stochastic differential equation, KDMS Conference, November 2022.
27. Irregularly sampled time series classification using neural stochastic differential equation, KIIE Annual Fall Conference, 2022.
28. Deep learning approach for behavior of piston of a linear compressor, KIIE Annual Fall Conference, 2022.
29. Domain knowledge-informed functional outlier detection for line quality control systems, KIIE Annual Spring Conference, 2022.
30. Multivariate time series classification using multi channel CNN classifier, KIIE Annual Spring Conference, 2022.
31. TAED-Net: Near ophthalmologist-level thyroid-associated eye symptom detection on frontal eye photographs with deep learning, KIIE Annual Spring Conference, 2022.
32. Risk assessment of abnormal vessel behaviours from AIS data using elastic depths, KIIE Annual Spring Conference, 2022.
33. (**Invited Talk**) Mixture gas classification and sensor drift compensation, LG H&A-UNIST Future Webinar 21, September 2021.
34. Commercial area analysis using big data on GIS: case of YongIn-si, KIIE Annual Spring Conference, 2021.
35. Sensor drift compensation for mixed gas classification under batch experiments, KIIE Annual Spring Conference, 2021.

36. Transfer-learning based approach for mixture gas classification, KIIE Annual Fall Conference, 2020.
37. Congestion propagation modeling with graph neural network (GNN), KIIE Annual Fall Conference, 2020.
38. Simulation-based anomaly detection in nuclear reactors, KIIE Annual Fall Conference, 2020.
39. Exploiting logistics anomaly detection using maritime big data, KIIE Annual Fall Conference, 2019.
40. Ordinal-imbalanced data classification through noise reduction by singular value decomposing truncation, KIIE Annual Spring Conference, 2019.
41. (**Invited Talk**) Data analytics in logistics systems: monitoring, classification, and assessment, Yonsei University, May 2018.
42. Sensor drift compensation using temperature and humidity for mixture gases classification, KIIE Annual Spring Conference, 2018.
43. (**Invited Talk**) Big data analytics in logistics, 4th UNIST Big Data Symposium, November 2017.

SERVICE

Campus Contributions

- Program Director of Industrial Safety AI, UNIST Novatus Graduate School, 2026-present
- Head, Department of Industrial Engineering, 2024-present
- Member, the Faculty Personnel Committee, Artificial Intelligence Graduate School, 2025-present
- Member, UNIST Artificial Intelligence Committee, 2025-2026
- Group Chair, The Industrial Cooperation Working Group, UNIST-LG Electronics, 2023-present
- Member, UNIST Academic Steering Committee, 2022-2024
- Chair, The Industrial Cooperation Committee, Artificial Intelligence Graduate School, 2021-2024
- Chair, The Faculty Recruitment Committee, Department of Industrial Engineering, 2021-2022
- Member, The Faculty Recruitment Committee, Department of Industrial Engineering, 2023
- Chair, The Department Education Committee, Department of Industrial Engineering, 2020-2023
- Member, UNIST-KAIST-POSTECH Data Science Competition Founding Committee, 2021
- Member, The University Admission Interview Committee, 2020
- Member, Election Commission for Faculty Representatives of the UNIST Council, 2020
- Member, The Fourth Industrial Revolution Working Committee, 2018-2022
- Member, UNIST Visibility Committee, 2018-2020

- Member, Space Planning and Allocation Committee for Industry-University Convergence Campus, 2018-2019
- Member, The Faculty Recruitment Committee, School of Management Engineering, 2017-2020
- Member, The Undergraduate Admissions Committee, 2017
- Member, Committee for the Preparation of the Movement to Complex Campus, School of Management Engineering, 2017

Professional Membership

- Senior Member, The Institute for Operations Research and the Management Sciences (INFORMS), 2024-present
- Member, The Institute for Operations Research and the Management Sciences, 2006-2024
- Member, The Institute of Industrial & Systems Engineers, 2019-present
- Member, The Korean Institute of Industrial Engineers, 2014-present

Public and Community Service

- Outside Director, Kakao Impact Foundation, 2026-2028
- Member, Logistics Policy Committee, Ulsan Metropolitan City Hall, 2019-2021, 2023-2024

Professional Activities

- Chair, Organizing Committee, 2026 Korea Data Mining Society (KDMS) Annual Summer Conference, August 2026
- Organizing Committee, APIEMS (Asia Pacific Industrial Engineering and Management Systems Conference), November 2026
- **Area Editor in Statistics, Quality, Reliability & Maintenance**, *Computers & Industrial Engineering*, 2023-present
- **Associate Editor**, *Journal of the Korean Institute of Industrial Engineers*, 2024-present
- Director, Korea Data Mining Society (KDMS), 2022-2026
- Board of Directors, IISE QCRE Division, 2025-2026
- Special session chair at IEEE CASE 2026, *AI for Safety, Sustainability, and Automation in Smart Maritime Transport and Logistics Systems*, Shenyang, China, August 2026
- QCRE Track Program Chair, *IISE Annual Conference & Expo 2025*, 2024-2025
- Session chair of the KIIE Annual Conference, ML & AI, Daejeon, November 2025
- Member (Overall/Planning), Organizing Committee, the KIIE Annual Conference, Fall 2023
- Session chair of the KIIE Annual Conference, Business Analytics, Jeju, June 2022
- Invited reviewer, The Faculty Recruitment Committee, Department of Industrial & Systems Engineering, KAIST, 2021-2022
- Session chair of the INFORMS Annual Meeting, VTA08. Data Mining II, October 2021
- Reviewer of the Mid-Career Researcher Program, National Research Foundation of Korea, 2021

- Session chair of the KIIE Annual Conference, Probability/Statistics/Quality, Jeju, June 2021
- Reviewer, IISE Transactions, IEEE Transactions on Automation Science and Engineering, Annals of Operations Research, 2021-present
- Session chair of the Fifth International Conference on the Interface between Statistics and Engineering (ICISE), Statistics and Analytics, Seoul, June 2019
- Session chair of the KIIE Annual Conference, Industrial AI, Gwangju, April 2019

PROFESSIONAL EXPERIENCE

Samsung SDS, Seoul, Korea
Data Analytics Lab, Algorithm Research Team, R&D Center
Senior Engineer/Data Scientist January 2014-June 2016

- Developed risk assessment and scoring algorithm using text mining for SDS Smart Logistics portal solution, Cello Square.
- Developed a data-driven method for early detection of vessel delays combined with real-time vessel tracking information. The proposed approach is validated by applying real datasets extracted from the logistics platform of Samsung SDS, Cello.
- Analyzed global IT trends and disruptive technologies in the fields of big data analytics, the Internet of Things (IoT), and video analytics.
- Collaborated with SDS R&D center in San Jose to research new technologies in data analytics for SDS mid/long term business in global markets.

Terra Technology, Norwalk, Connecticut, USA
Supply Chain Management Consultant September 2011-December 2013

- Applied data mining techniques (e.g., pattern recognition, regression analysis, clustering) to real business problems.
- Collected/identified big data from existing internal databases and external/public data sources.
- Cooperated with the client's DBA team and tested the performance of forecasts.
- Developed technical solutions, including writing and testing SQL procedures and/or UNIX/Windows shell scripts to meet customer integration requirements.
- Participated in demand planning projects for Procter & Gamble, Unilever, Kraft Foods Inc., ConAgra Foods, Kellogg's, etc.
- Analyzed data to identify demand volatility and changing consumer preferences.
- Created data validation templates using Qlikview reports by adding new functionalities: multiple selections, high dimensional search.

Predictix, Atlanta, Georgia, USA
Data Scientist Intern May-December 2009 & May-August 2008

- Participated in pre-sales projects for Target Corporation.
- Developed a synthetic sales data generator for studying promotion effects.
- Analyzed retail sales transaction data for promotion planning.

- Developed spatial-temporal data mining techniques for retail sales transaction data from Shopko.
- Collected/identified big data from existing internal databases and external/public data sources.

Georgia Institute of Technology, Atlanta, Georgia, USA
H. Milton Stewart School of Industrial and Systems Engineering
Research Assistant May 2006-August 2011

- Developed statistical methodologies for a data collection plan with uncertain design space.
- Developed a new approach, *Layers of Experiments*, for the robust optimization of nanoparticle synthesis.
- Explored a new statistical methodology in a multi-level, multi-scale framework in the context of supply chain logistics systems.
- Performed inventory system analysis on stocked items and participated in a warehousing design for Enraf Fluid Technology.

TECHNICAL SKILLS

R, Python, Julia, MATLAB, Hadoop, Spark, Perl, C/C++, Korn Shell, Unix/Linux, Java, SQL, MS Access, PL/SQL, XML, MySQL, LINDO/LINGO, GAMS, Qlikview

AWARDS & HONORS

- 2024 IISE Best Paper Competition Finalist, Data Analytics and Information Systems Division, *The Institute of Industrial and Systems Engineers*
- 2023 Excellence in Mentoring Award, *College of Information and Biotechnology, UNIST*
- 2023 KCC Best Presentation Award, AI Division, *Korean Institute of Information Scientists and Engineers*
- 2023 KCC Best Paper Award, AI Division, *Korean Institute of Information Scientists and Engineers*
- 2023 IISE Best Paper Competition Finalist, Data Analytics and Information Systems Division, *The Institute of Industrial and Systems Engineers*
- 2021 IISE Best Paper Award, Logistics and Supply Chain Division, *The Institute of Industrial and Systems Engineers*
- 2019 IISE Best Paper Award, Quality Control & Reliability Engineering Division, *The Institute of Industrial and Systems Engineers*
- 2015 SDSers of the Quarter, *Samsung SDS*
- 2010 Honorable mentions, *The Business and Economics Statistics Section at the Joint Statistical Meetings*
- 2006 Global Logistics Scholar, *The Logistics Institute, Georgia Institute of Technology*
- 2005 Graduate Research Scholarship, *Korea Research Foundation*